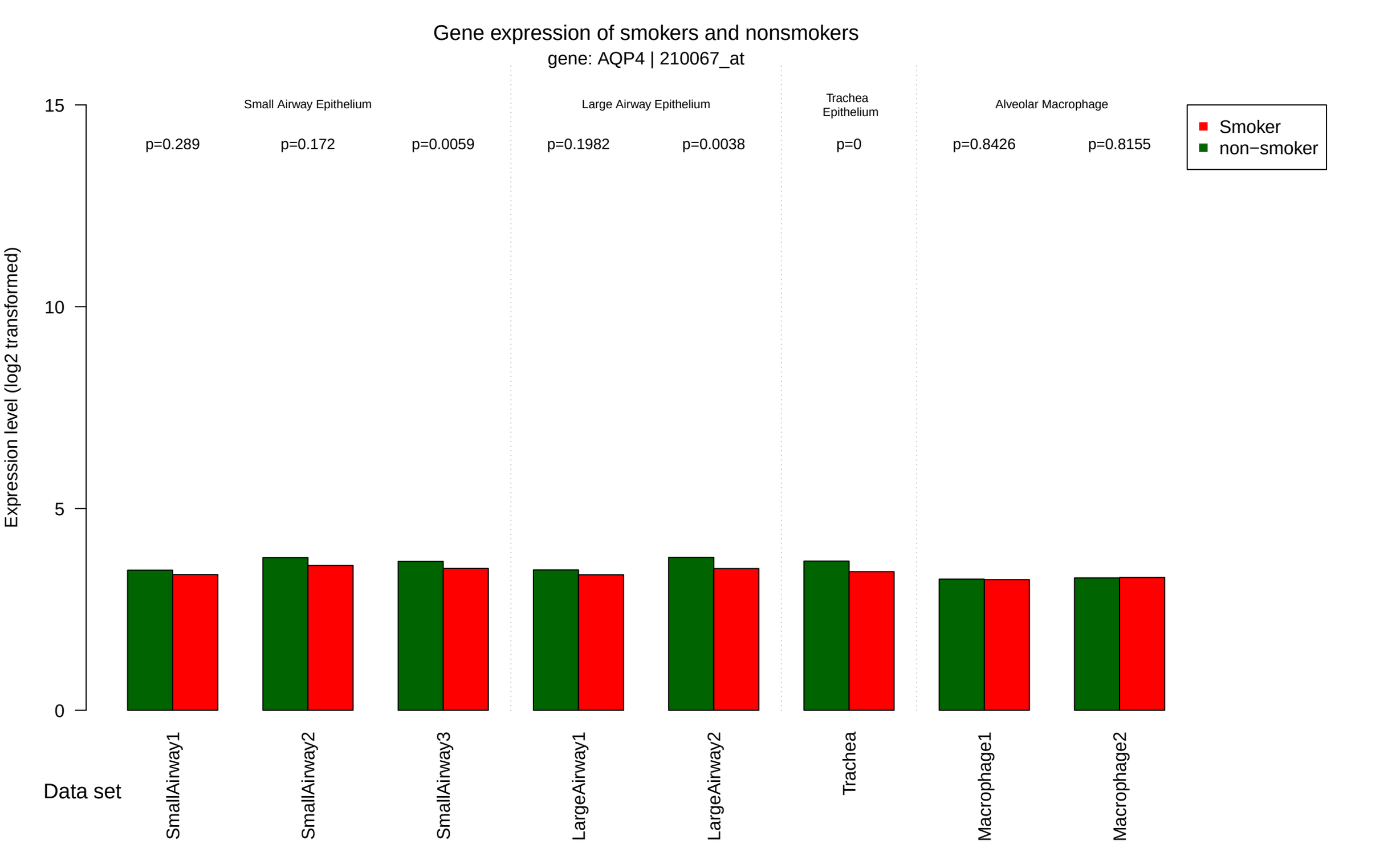
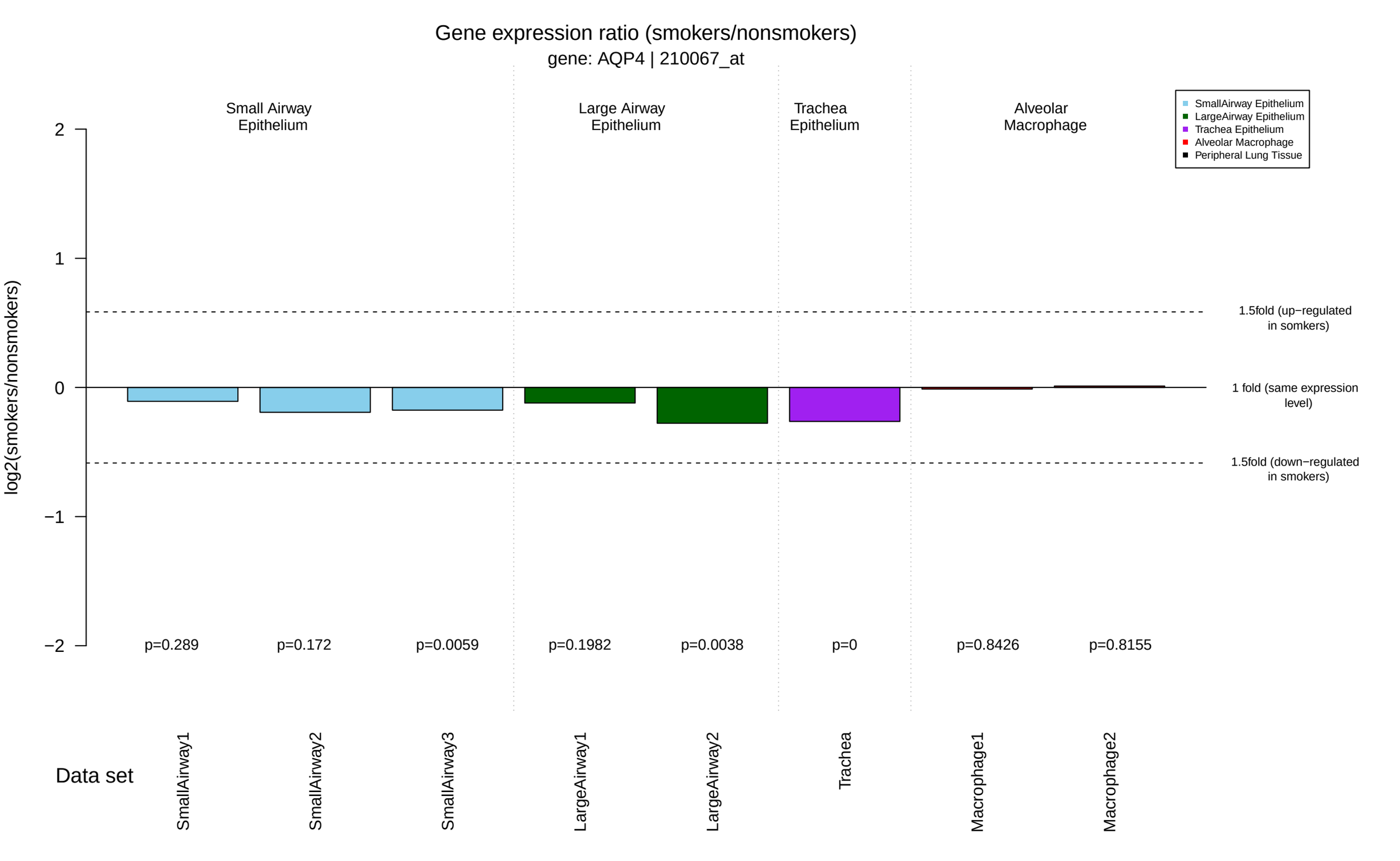
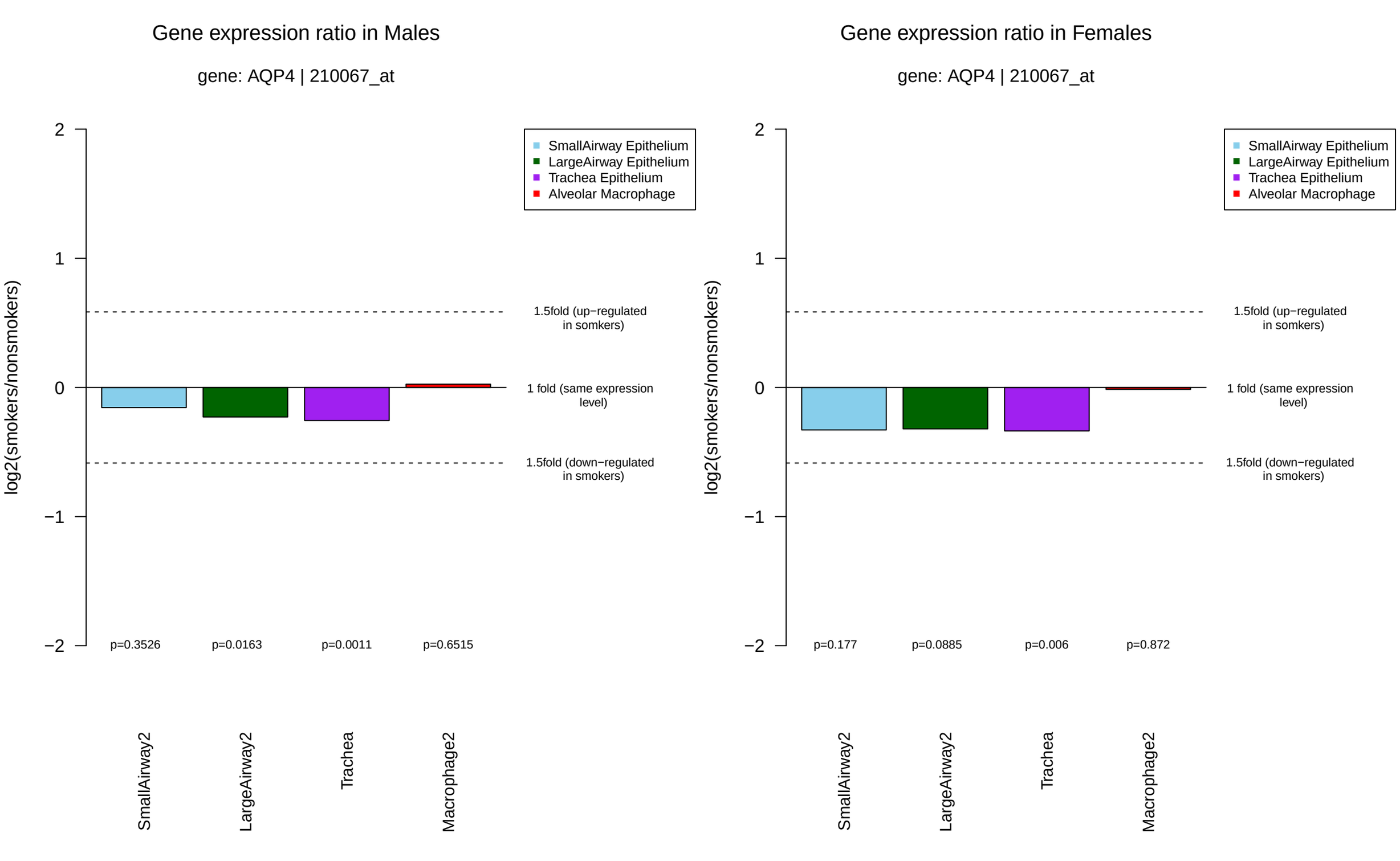
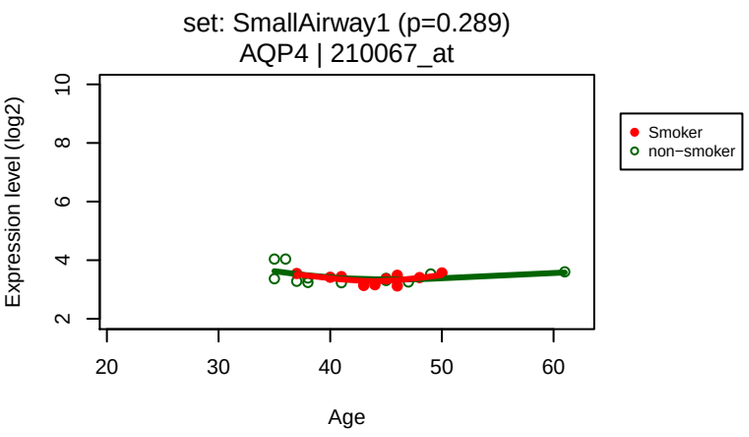
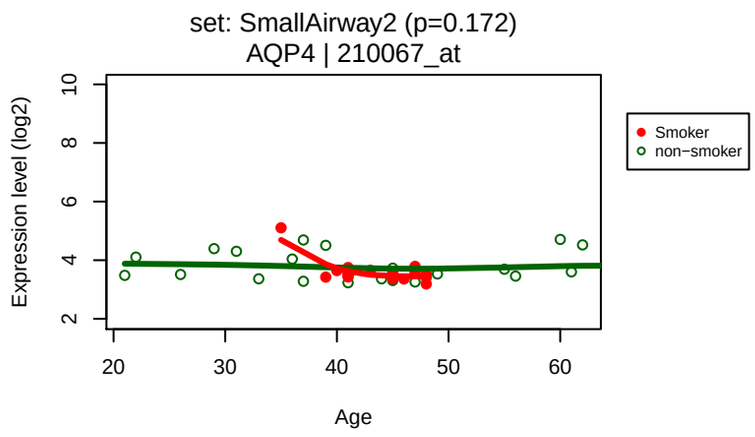
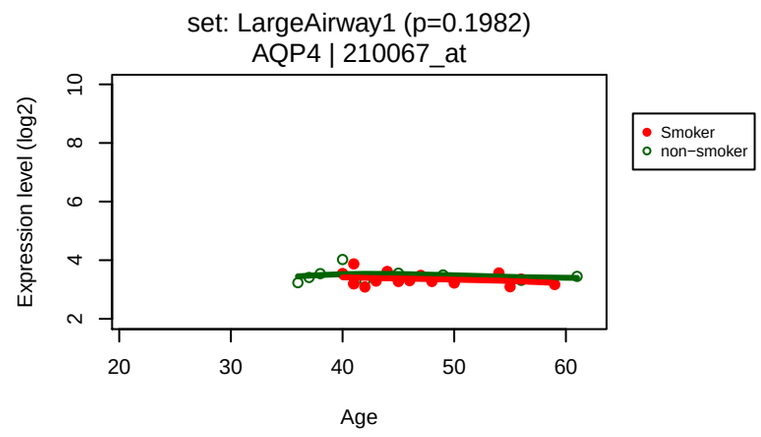

Gene expression of smokers and nonsmokers
gene: AQP4 | 210067_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.289
p=0.172
p=0.0059
p=0.1982
p=0.0038
p=0
p=0.8426
p=0.8155
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

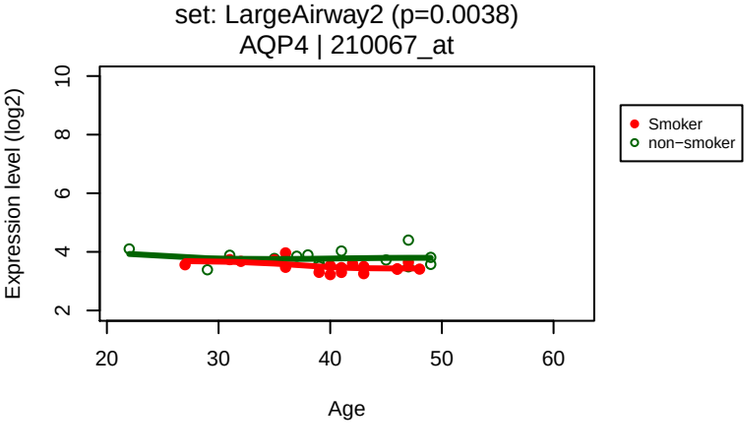

Gene expression ratio (smokers/nonsmokers)
gene: AQP4 | 210067_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.289
p=0.172
p=0.0059
p=0.1982
p=0.0038
p=0
p=0.8426
p=0.8155
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

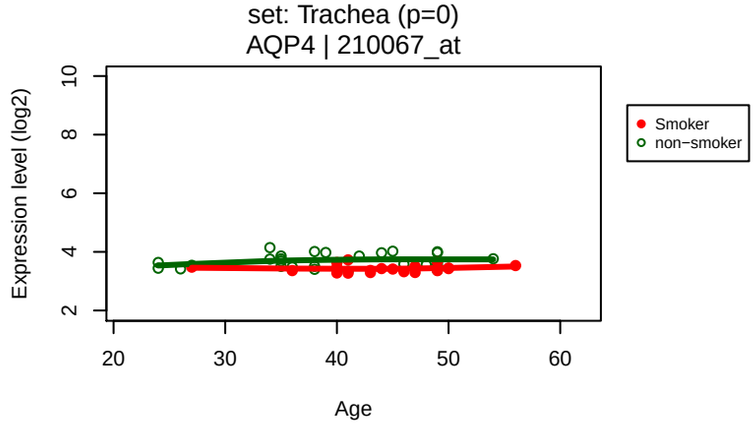

Gene expression ratio in Males
gene: AQP4 | 210067_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3526
p=0.0163
p=0.0011
p=0.6515
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AQP4 | 210067_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.177
p=0.0885
p=0.006
p=0.872
SmallAirway2
LargeAirway2
Trachea
Macrophage2

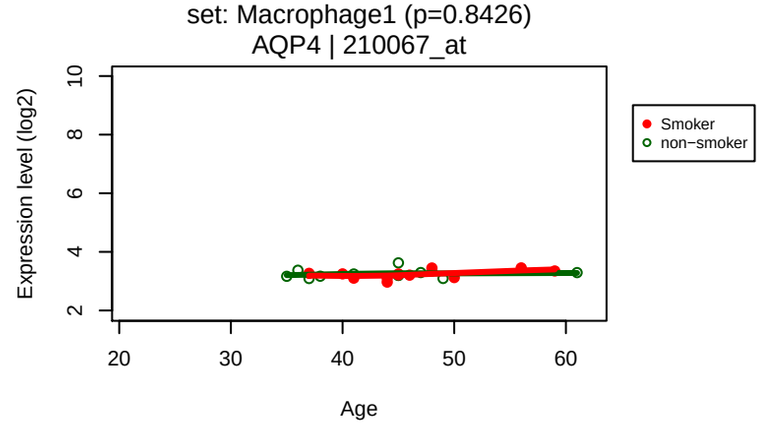

set: SmallAirway1 (p=0.289)
AQP4 | 210067_at
Expression level (log2)
Age
Smoker
non−smoker

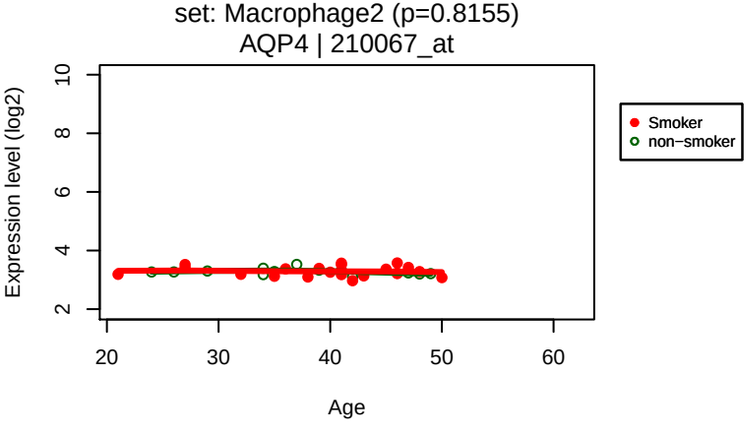

set: SmallAirway2 (p=0.172)
AQP4 | 210067_at
Expression level (log2)
Age
Smoker
non−smoker

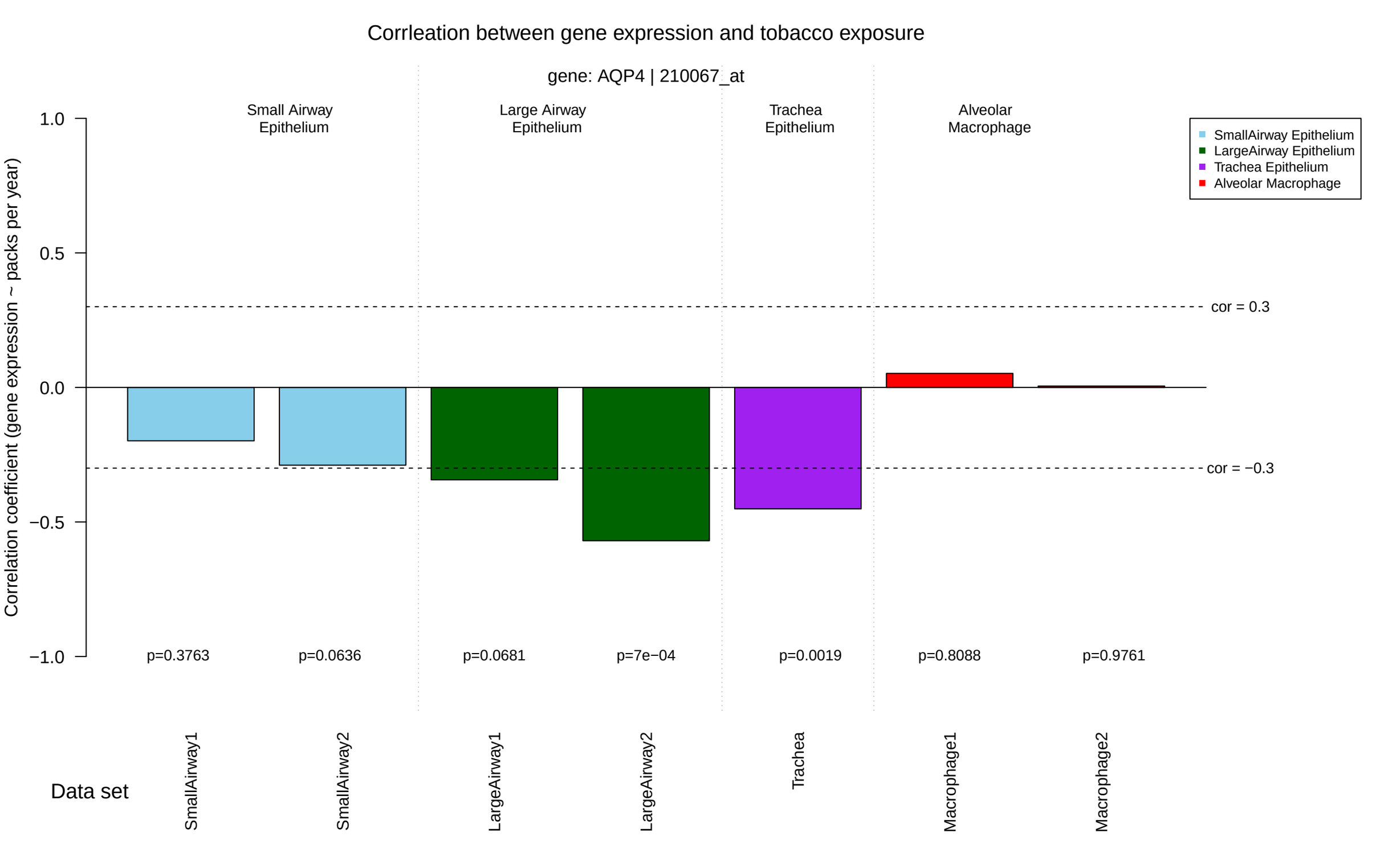

set: LargeAirway1 (p=0.1982)
AQP4 | 210067_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0038)
AQP4 | 210067_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0)
AQP4 | 210067_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8426)
AQP4 | 210067_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8155)
AQP4 | 210067_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AQP4 | 210067_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3763
p=0.0636
p=0.0681
p=7e−04
p=0.0019
p=0.8088
p=0.9761
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2